

CY201

Cyber Security Principles and Concepts

Semester Project Guidelines

Spring 2026

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These guidelines are fully aligned with the official course syllabus, the three Course Learning Outcomes (CLOs), and the published assessment structure. The project is worth **10% of the overall course grade** and forms a major part of the “Assignments/Project” component.

These guidelines are designed using current **industry standards and best practices** (NIST Cybersecurity Framework, OWASP Top 10, ISO 27001, CIS Controls, and Zero-Trust principles). The project converts theoretical knowledge from all 15 weeks of the syllabus into a **portfolio-ready, hands-on deliverable** that demonstrates your ability to analyze risks, apply security controls, and present professional cybersecurity solutions.

1 Project Objectives (Directly Mapped to CLOs)

By completing this project, you will achieve the following measurable outcomes:

- **CLO-1 (Understanding – 20% of project marks):** Demonstrate a clear grasp of the fundamental principles of cybersecurity across network, web, mobile, cloud, and emerging domains.
- **CLO-2 (Conceptual Analysis – 30% of project marks):** Use cybersecurity conceptual knowledge to analyze threats, vulnerabilities, and required security measures.
- **CLO-3 (Applying – 50% of project marks):** Apply a profound comprehension of cybersecurity principles and concepts through practical design, implementation, and evaluation of real-world security solutions.

2 Project Scope & Mandatory Requirements

You must **analyze, design, simulate/implement, evaluate, and professionally document** a complete cybersecurity solution that integrates concepts from **every major topic** in the syllabus (Weeks 1–15).

2.1 Core Technical Requirements (All Must Be Demonstrated)

- **Threat Modeling & Risk Analysis:** Identify assets, threats, vulnerabilities, and attack vectors (social engineering, web, IoT, cloud).
- **Cryptography & Encryption:** Implement at least two cryptographic primitives (symmetric/asymmetric, hashing, digital signatures) and demonstrate secure key management or VPN/SSL/TLS usage.
- **Network/Web/Mobile/Cloud Security:** Apply controls for web application security (OWASP Top 10), operating-system hardening, mobile device management, and cloud security best practices.
- **Identity & Access Management:** Design authentication, authorization, and multi-factor mechanisms.
- **Incident Response & Forensics:** Develop an incident response plan and show basic digital forensics steps.
- **Security Policies & Compliance:** Draft a realistic organizational cybersecurity policy, including risk management, regulatory compliance, and audits.
- **Emerging Trends:** Address at least one modern area (IoT/embedded security, threat intelligence, or Zero-Trust Architecture).
- **Performance & Validation:** Use tools to simulate attacks/defenses and measure effectiveness.

2.2 Approved Tools (Industry-Standard & Freely Available)

- Kali Linux (VM), Wireshark, Burp Suite Community, OWASP ZAP, OpenSSL.
- VirtualBox/VMware, Metasploit (community), Autopsy or Volatility.
- Diagramming: Draw.io or Lucidchart (professional threat models).
- Optional (bonus-eligible): Python scripting, NIST-compliant policy templates.

2.3 Group Size & Project Themes

Group Size: 3–4 students (strongly recommended). Individual projects require prior written approval.

Project Themes (Choose One or Propose Your Own — Approval Required):

1. Comprehensive Cybersecurity Policy & Implementation for a Hypothetical Enterprise.
2. Vulnerability Assessment & Penetration Testing of a Sample Web/Mobile Application.
3. Secure Cloud & IoT Deployment with Threat Modeling and Zero-Trust Controls.
4. Incident Response Plan & Digital Forensics Investigation Simulation.
5. Cryptography & Network Security Solution for a Multi-Device Environment.
6. Cyber Threat Intelligence Dashboard + Security Awareness Training Module.

3 Deliverables & Timeline (Milestones)

All submissions are to be made through the course Moodle page. Late submissions incur a **10% penalty per day**.

- **Milestone 1: Project Proposal** — 1–2 page document including project title, group members, chosen theme, high-level threat model diagram, mapping to syllabus weeks, and tools to be used.
- **Milestone 2: Mid-Semester Progress Report & Demo** — Partial implementation, threat analysis summary, and a short 5-minute in-class presentation/demo.
- **Milestone 3: Final Submission & Presentation**
 - **Comprehensive Final Report** (15–20 pages, IEEE formatting style) covering executive summary, detailed threat model, design decisions with diagrams and configuration/code snippets, analysis of security controls, simulation results, policy document, limitations, lessons learned, and references.
 - **Live Demonstration or Recorded Video** (10–15 minutes): working security implementation plus realistic attack/defense or forensics scenario.
 - **Presentation Slides** (12–15 slides): problem statement, methodology, key findings, results, and reflections.
 - **Complete Source Package**: lab files, scripts, packet captures, policy templates, and raw evidence.
 - **Peer Evaluation Form**: confidential individual contribution assessment.

4 Submission Template Support

Your submission-ready Word template is ready:

→ **Download the CY201 Project Template** (available on Moodle under the Project section).

To make your report look professional and save you time, the instructor can also:

- Format it with a cover page, automatic table of contents, and full GIK Institute branding
- Convert the completed report into a clean PDF version for final submission
- Add sample diagrams and placeholder figures (threat models, Wireshark screenshots, policy templates, etc.) that you can simply replace with your own work

Using the template is **strongly recommended** as it ensures consistency, meets academic and industry documentation standards, and helps you focus on technical content rather than formatting.

5 Evaluation Rubric (Transparent & Aligned with CLOs)

6 Policies & Student Support

- **Academic Integrity**: Zero tolerance for plagiarism (Turnitin check). Cite all sources properly.
- **Attendance**: 80% mandatory to appear in the final examination (institute policy).

Criterion	Weight
Technical Implementation & Correctness	35%
Analysis & Threat Modeling	30%
Security Policies & Best Practices	15%
Documentation Quality & Professionalism	10%
Presentation & Demo Quality	10%

- **Collaboration:** Every group member must fully understand and be able to explain all parts of the project.
- **Support Resources:** Stallings “Network Security Essentials” (7th Ed.), Andress “The Basics of Information Security” (3rd Ed.), Sandhu & Boles “Cybersecurity: Principles and Practice” (3rd Ed.); OWASP, NIST Cybersecurity Framework, Kali Linux documentation.
- **Office Hours:** 11:00 am – 2:00 pm (S-02 FCSE) — use them early and often.
- **Bonus Opportunities:** Up to 5% extra credit for projects incorporating advanced elements (live CTF-style demo, AI-assisted threat intelligence, ELK stack, etc.).

This project is your opportunity to build a **realistic industry-grade cybersecurity solution** and a professional portfolio piece that you can proudly showcase in job applications, internships, or technical interviews.

For template access, custom project approval, or any clarification, please contact the instructor during office hours.

Approved by:

Engr. Muhammad Ahmad Nawaz
Instructor, CY201
April 2026